



SGGS Institute of Engineering & Technology, Nanded
Department of Electronics & Telecom. Engg.
END-SEMESTER Examination
B.Tech. (EXTC)

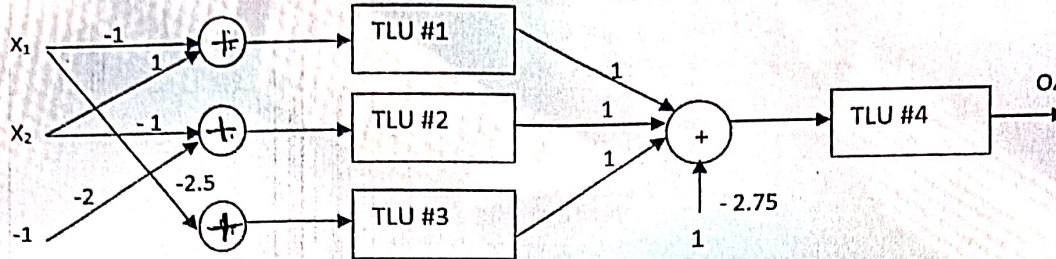
Artificial Intelligence and Neural Networks OEC-EC407

Date: 19 Nov. 2025 (13:30-15:30)

Max marks: 60

Note: I. Attempt ANY FIVE questions
II. Assume suitable data if necessary

Q.1 A FEEDFORWARD network is shown in the following figure. Find the segment of the x_1, x_2 plane for which the output $O_4=1$. (12)



Q.2 What is EXOR problem in classification? Give a possible solution using a Single layer Feed-forward neural network. (12)

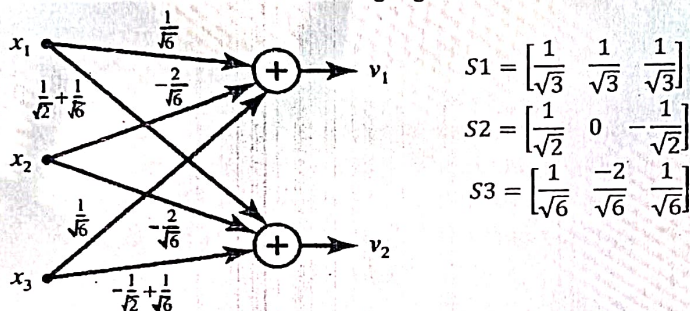
Q.3 Write the proof for Perceptron Convergence theorem. (12)

Q.4 Design an Associative Memory to store following vector (representing some TEXT) associations given by (12)

Association Pair Number	A	B
1	$[1 \ -1 \ -1 \ 1 \ -1]^T$	$[1 \ 1 \ -1 \ -1 \ 1]^T$
2	$[1 \ 1 \ -1 \ -1 \ 1]^T$	$[1 \ -1 \ 1 \ -1 \ 1]^T$
3	$[1 \ -1 \ 1 \ -1 \ 1]^T$	$[1 \ -1 \ -1 \ 1 \ -1]^T$

Compute the output of the network if; $[1 \ -1 \ 1 \ -1 \ 1]^T$ is applied as stimulus.

Q.5 A linear associator is shown in the following Figure:



Find the weight matrix W and then compute Outputs $F1, F2, F3$ for each Stimulus given above. (12)

Q.6. The weight matrix W for a single layer feedback neural network with bipolar discrete binary neurons is given as

$$W = \begin{bmatrix} 0 & -1 & 1 & -1 \\ -1 & 0 & 1 & -1 \\ 1 & 1 & 0 & -1 \\ -1 & -1 & -1 & 0 \end{bmatrix}$$

If thresholds and external inputs of neurons are zero, compute energy equation and different energy levels associated with the network. (12)

Course Outcomes (COs):

1. Understanding the basics of Artificial Intelligence, artificial neural network, their limitations, basic pattern analysis tasks such as classification and clustering, learning and adaptation using the learning rules, implementation of learning rule.
2. Describe the concepts of feed forward neural networks using single layer and multilayer networks to solve classification problem, and its implementation, single layer feedback networks to study the concept of memory using neural networks.
3. Analyze and implement the applications of neural networks in character recognition and control systems.
4. Understand Auto associative neural networks, Pattern storage and retrieval, Hopfield model, recurrent neural networks

Bloom's Taxonomy (BT)

BT1- Remember. BT2- Understand. BT3- Apply. BT4- Analyse. BT5- Evaluate. BT6- Create

Q. No.	1	2	3	4	5	6
CO No.	2	2	1	3	4	4
BT No.	2	1	2	3	5	4

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Electronics and Telecommunication End Term Examination Academic Year: 2025-26 (Semester-I)	
Class: Final Year B.Tech.	Course: Data Communication and Networking	Course Code: PEC-EC-403
Date: 22.11.2025	Time: 13:30 to 15:30	Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	All Questions are compulsory	Marks	CO	BT
Q.1	Draw and explain the TCP/IP and OSI reference models in detail.	10	CO1	BT1
Q.2	Write short note on: i) Infrared Transmission ii) Microwave Transmission	10	CO3	BT2
Q.3	Station A uses 32-byte packets to transmit message to station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use?	10	CO5	BT5
Q.4	Explain Transmission Control Protocol (TCP) Segment Header in detail.	10	CO2	BT3
Q.5	Explain CSMA/CD protocol in detail.	10	CO3	BT2
Q.6	Explain the following: i) World Wide Web ii) Multimedia	10	CO2	BT2

== ALL THE BEST ==



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electronics and Telecommunication Engineering
Academic Year: 2025-26 (semester I)
End-Semester Examination

Class: B.Tech.

Course Name: Microwave Engineering

Course Code: PCC-EC-401

Date: 25.11.2025

Time: 13.30 to 15.30

Max. Marks: 60

Note:

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- Figures to right indicate full marks.
- The symbols carry usual meaning.
- Solve any six questions.

Q.N.			Marks	CO	BT
Q.1	a	A rectangular cavity (vacuum inside) has internal dimensions $a=4.0$ cm (x-dimension) $b=2.0$ cm (y-dimension) $d=6.0$ cm (z-dimension) Find the resonant frequency of the TE_{101} mode.	5	CO1	BT5
	b	A 3-port circulator is non-ideal with the following measured characteristics at an operating frequency: insertion loss (port $\rightarrow$ next port) = 1.0 dB, isolation (port $\rightarrow$ isolated port) = 28 dB. Assume matched ports and negligible return loss. If $P_{in} = +20$ dBm is fed into port 1, find the power (in dBm and mW) at: port 2 (desired port), and port 3 (isolated port).	5	CO2	BT4
Q.2	a	Explain various types of circulators.	5	CO3	BT1
	b	Explain various types of waveguide cavities.	5	CO4	BT2
Q.3	a	A load shows VSWR = 1.5 when driven by 1 W of forward power (i.e. 30 dBm). An isolator (isolation 20 dB) sits between generator and load. Compute: i) reflection coefficient magnitude $ \Gamma $, ii) reflected power at the load (mW), iii) reflected power that reaches the generator after the isolator (mW and dBm)	5	CO5	BT4
	b	Explain the internal structure of TRAPATT diode in detail.	5	CO6	BT6
Q.4	a	Explain the working principle, construction and application of isolator in detail.	5	CO7	BT2
	b	Explain the structure and working of Travelling Wave Tube in detail.	5	CO8	BT3
Q.5	a	Explain micro strip line in detail.	5	CO9	BT2

	b	Explain with neat diagram magnetron oscillator.	5	CO10	BT2
Q.6	a	Design the strip width W for a 50-Ω micro strip Given: substrate relative permittivity $\epsilon_r = 4.4$, substrate thickness $h = 1.6$ mm. Target characteristic impedance $Z_0 = 50 \Omega$.	5	CO11	BT3
	b	Explain the structure and working of two cavity klystron amplifier.	5	CO12	BT2
Q.7	a	Explain Gunn diode in detail.	5	CO13	BT2
	b	Explain the structure and working of reflex klystron amplifier.	5	CO14	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Calculate resonant frequency for TE ₁₀₁ modes in rectangular cavity
CO2	Solve numerical problems related to circulator parameters such as insertion loss, isolation, return loss, and power flow.
CO 2	Calculate power at various ports of circulator
CO 3	Explain the working principle and function of microwave circulators used for signal routing in microwave systems.
CO 4	Explain the working principle and function of waveguide cavities in microwave systems.
CO 5	Calculate various performance parameters of isolators
CO 6	Understand the basic structure and operation of the TRAPATT diode.
CO7	Explain the working principle, construction and application of isolator
CO 8	Explain the working principle, construction, and operation of a Traveling Wave Tube (TWT) and its role as a microwave amplifier.
CO9	<i>Explain the structure, working principle, and characteristics of a micro strip transmission line used in microwave circuits.</i>
CO10	Explain the construction, working principle, and operation of a magnetron oscillator used for microwave generation.
CO11	Analyze impedance and dielectric properties of micro strip line using numerical data
CO12	Understand the construction and components of a two-cavity klystron, including the buncher and catcher cavities.
CO13	Explain the principle of negative differential resistance (Gunn effect) responsible for microwave generation.
CO14	Understand the structure and main components of a reflex klystron, including the cavity, repeller electrode, and electron beam system.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

End Term Examination (Semester-I) 2025-26

B. Tech. (Final Year)

Data Mining and Data Warehousing

Electronics and Telecommunication Engineering

OEC-EC418

Date: 20/11/2025

Time: 13:30 to 15:30

Max. Marks: 60

- Note:
- Attempt all questions.
 - Assume suitable data if necessary.
 - Use of non-programmable calculator is permitted.

Q No	Question	Marks	CO	BT
Q. 1	Explain the different methods of indexing OLAP data.	12	OEC-EC-418-3	BT2
Q. 2	Design similarity and dissimilarity between the objects. Find SMC and Jaccard's coefficient of two binary vectors. $X = (1,0,0,0,0,0,0,0,0)$ $Y = (0,0,0,0,0,0,1,0,0,1)$	12	OEC-EC-418-2	BT5, 6
Q. 3	State Apriori principle. Write apriori algorithm for frequent itemset.	12	OEC-EC-418-3	BT4
Q. 4	Explain different types of clusters.	12	OEC-EC-418-4	BT5
Q. 5	a. State and explain K-means algorithm.	06	OEC-EC-418-1	BT5
	b. Write the characteristics of nearest neighbour classifier.	06	OEC-EC-418-2	BT2

Course Outcomes (CO)

OEC-EC-418-1: Discuss basic concepts of Data Warehouse fundamentals, data analytics, data mining using professional language associated with data analytics and data mining.

OEC-EC-418-2: Design data warehouse with dimensional modelling and apply OLAP operations.

OEC-EC-418-3: Pre-process the data so that it can be analysed further using sophisticated data analytics and mining algorithms.

OEC-EC-418-4: Identify appropriate data mining algorithms to solve real world problems.

OEC-EC-418-5: Benefit the user experiences towards research and innovation integration.

Bloom's Taxonomy (BT)

BT1- Remember

BT2- Understand

BT3- Apply

BT4- Analyze

BT5- Evaluate

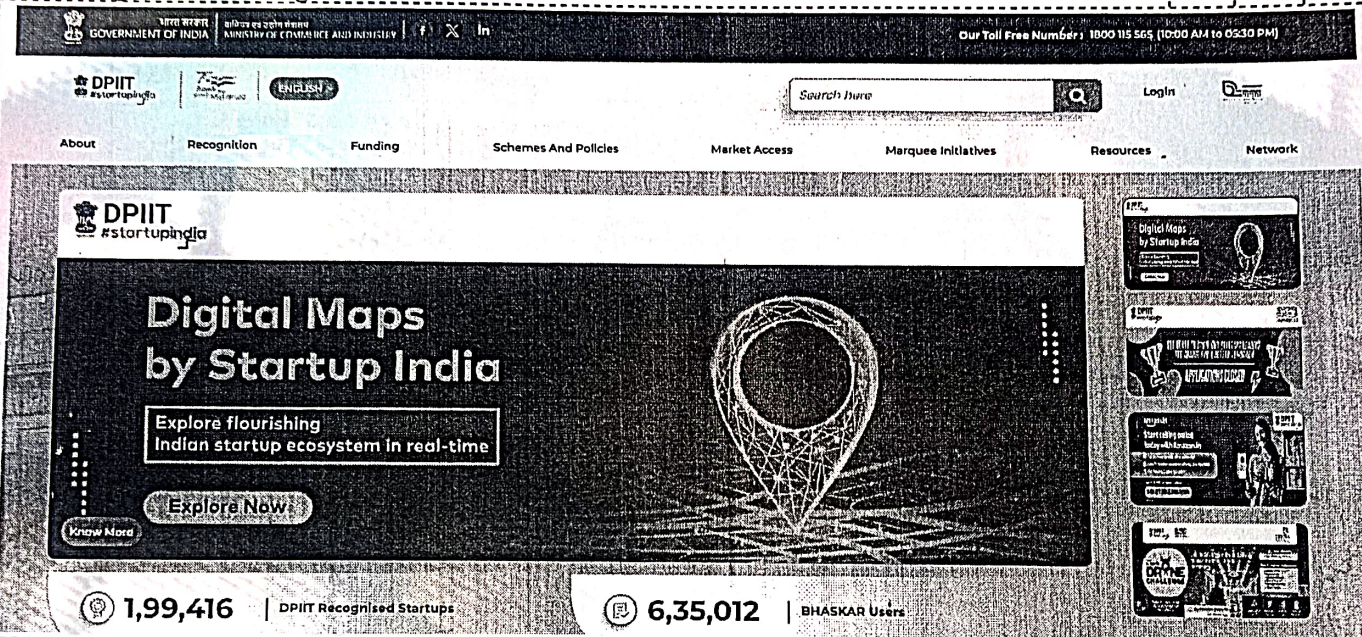
BT6- Create

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded	
	END TERM EXAMINATION 2025-26 (Semester-I/ VII)	
	Final Year B. Tech.	Electronics and Telecommunication Engineering
	Entrepreneurial Economics - EE	IIMC - EC409
Date: 27/Nov/2025 (Thur)	Time: 13:30 to 15:30	Max. Marks : 20

Note: i. Assume suitable data if necessary. ii. Use of non-programmable calculators is permitted.
iii. This paper contains one page only. iv. Solve all questions.

Q.N	Question	Marks	CO	BT
Q.1	Explain how economic freedom contributes to the overall prosperity of a nation.	10	5	1,2,4
Q.2	Discuss how profit and loss serve as indicators of market performance.	10	2	1,2,5
Q.3	What is meant by the term " marvel of the market "? Describe how it operates in practice.	10	1	1,2,4
Q.4	Write a detailed note on your favorite unicorn. Mention steps to register a startup.		2,4,6	2,3,4
Q.5	Write the correct url for the web page shown in figure 5. List all the tabs you see on the page. Write full form of DPIIT. Write significance of the digits 1,99,416 and 6,35,012	10	1,2,4	1,2

Figure 5: Web Page



Q.5	Match the items listed in Column I with their corresponding entries in Column II. Rewrite the correctly matched pairs in your answer sheet. For each pair, provide a brief explanatory note highlighting the connection or significance of the match.		10	1,2,3,4	1,2
	Column I	Column II			
	IPIndia	https://www.startupindia.gov.in/			
	Startup India	https://www.mca.gov.in/content/mca/global/en/home.html?trk=article-ssr-frontend-pulse			
	Ministry of Corporate Affairs	https://ipindia.gov.in/			

*****!!!!!!All the Best!!!!!!*****

Details below are for Information only: **Bloom's Taxonomy:** BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create **Course outcomes:** 1. Define entrepreneurship 2. Identify examples of entrepreneurship 3. Articulate the traits of an entrepreneur 4. Describe the importance of the entrepreneur's role in society 5. Provide examples of how to serve needs in their own community through entrepreneurial action



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electronics and Telecommunication
End Term Examination
Academic Year: 2025-26 (Semester-I)

Class: Final Year B.Tech.	Course: Data Communication and Networking	Course Code: PEC-EC-403
Date: 22.11.2025	Time: 13:30 to 15:30	Max. Marks: 60

Note:

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- v. The symbols carry usual meaning.

Q.N.	All Questions are compulsory	Marks	CO	BT
Q.1	Draw and explain the TCP/IP and OSI reference models in detail.	10	CO1	BT1
Q.2	Write short note on: i) Infrared Transmission ii) Microwave Transmission	10	CO3	BT2
Q.3	Station A uses 32-byte packets to transmit message to station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use?	10	CO5	BT5
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Q.6	Explain the following: i) World Wide Web ii) Multimedia	10	CO2	BT2

== ALL THE BEST ==

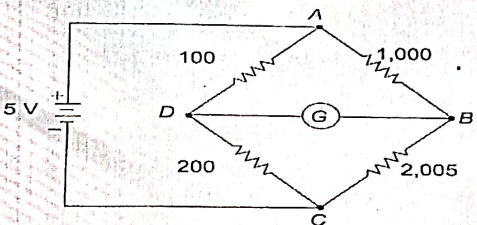


Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electronics and Telecommunication Engineering
Academic Year: 2025-26 (semester I)
End Term Examination

Class: Final Year B.Tech. Course Name: Electronic Instrumentation Course Code: PCC-EC402
Date: 20.11.2025 Time: 13:30 to 15:30 Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Figures to right indicate full marks.
- All questions are compulsory.

Que. No.	Question	Marks	CO	BT
Q.1	Ten measurements of the resistance of a resistor gave 101.2Ω , 101.7Ω , 101.3Ω , 101.0Ω , 101.5Ω , 101.3Ω , 101.2Ω , 101.4Ω , 101.3Ω and 101.1Ω . Assume that only random errors are present. Calculate (i) the arithmetic mean, (ii) the standard deviation of the readings and (iii) the probable error.	10	CO1	BT5
Q.2	Derive equation for the frequency of applied voltage in Wien Bridge.	10	CO1	BT4
Q.3	Derive equation for value of unknown resistor using Kelvin double bridge.	10	CO1	BT4
Q.4	Justify using circuit diagram and derivation of equation for the measurement of inductance of unknown inductor having high Q using Hay bridge does not need the frequency of voltage source.	10	CO1	BT4
Q.5	Prove that the light intensity measurement becomes simpler using photomultiplier than vacuum tube photo emissive cell.	10	CO2	BT4
Q.6	The following figure shows the schematic diagram of a Wheatstone bridge with values of the bridge elements as shown. The battery voltage is 5V and its internal resistance is negligible. The galvanometer has a current sensitivity of $10\text{mm}/\mu\text{A}$ and an internal resistance of 100Ω. Calculate the deflection of the galvanometer caused by the 5Ω unbalance in arm BC. 	10	CO3	BT5

== ALL THE BEST ==



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electronics and Telecommunication Engineering
Academic Year: 2025-26 (semester I)
End-Semester Examination

Class: B.Tech.

Course Name: Microwave
Engineering

Course Code: PCC-EC-401

Date: 25.11.2025

Time: 13.30 to 15.30

Max. Marks: 60

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	b	A 3-port circulator is non-ideal with the following measured characteristics at an operating frequency: insertion loss (port→next port) = 1.0 dB, isolation (port→isolated port) = 28 dB. Assume matched ports and negligible return loss. If $P_{in} = +20$ dBm is fed into port 1, find the power (in dBm and mW) at: port 2 (desired port), and port 3 (isolated port).	5	CO2	BT4
Q.2	a	Explain various types of circulators.	5	CO3	BT1
	b	Explain various types of waveguide cavities.	5	CO4	BT2
Q.3	a	A load shows VSWR = 1.5 when driven by 1 W of forward power (i.e. 30 dBm). An isolator (isolation 20 dB) sits between generator and load. Compute: i) reflection coefficient magnitude $ \Gamma $, ii) reflected power at the load (mW), iii) reflected power that reaches the generator after the isolator (mW and dBm)	5	CO5	BT4
	b	Explain the internal structure of TRAPATT diode in detail.	5	CO6	BT6
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	b	Explain the structure and working of Travelling Wave Tube in detail.	5	CO8	BT3
Q.5	a	Explain micro strip line in detail.	5	CO9	BT2

	b	Explain with neat diagram magnetron oscillator.	5	CO10	BT2
Q.6	a	Design the strip width W for a $50\text{-}\Omega$ micro strip Given: substrate relative permittivity $\epsilon_r = 4.4$, substrate thickness $h = 1.6\text{ mm}$. Target characteristic impedance $Z_0 = 50\text{ }\Omega$.	5	CO11	BT3
	b	Explain the structure and working of two cavity klystron amplifier.	5	CO12	BT2
Q.7	a	Explain Gunn diode in detail.	5	CO13	BT2
	b	Explain the structure and working of reflex klystron amplifier.	5	CO14	BT2

=== ALL THE BEST ===

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Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Calculate resonant frequency for TE 101 modes in rectangular cavity
CO2	Solve numerical problems related to circulator parameters such as insertion loss, isolation, return loss, and power flow.
CO 2	Calculate power at various ports of circulator
CO 3	Explain the working principle and function of microwave circulators used for signal routing in microwave systems.
CO 4	Explain the working principle and function of waveguide cavities in microwave systems.
CO 5	Calculate various performance parameters of isolators
CO 6	Understand the basic structure and operation of the TRAPATT diode.
CO7	Explain the working principle, construction and application of isolator
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